

| Question  | Answer                                                                                                                              | Marks     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 2(a)      | (thermal) energy per unit mass (to change temperature)                                                                              | <b>B1</b> |
|           | (thermal) energy per unit change in temperature                                                                                     | <b>B1</b> |
| 2(b)(i)   | work done = $p\Delta V$<br>$= (2.0 \times 10^5) \times (0.063 - 0.038) = 5000 \text{ J}$                                            | <b>A1</b> |
| 2(b)(ii)  | gas is expanding (against external pressure)                                                                                        | <b>B1</b> |
|           | gas does work / work is done by gas, so (work done on gas is) negative                                                              | <b>B1</b> |
| 2(b)(iii) | $\Delta U = q + W$                                                                                                                  | <b>C1</b> |
|           | $7600 = q + (-5000)$                                                                                                                | <b>A1</b> |
|           | $q = 12\,600 \text{ J}$                                                                                                             |           |
| 2(b)(iv)  | specific heat capacity = $q / m\Delta T$<br>$= 12600 / (0.35 \times 56)$                                                            | <b>C1</b> |
|           | $= 640 \text{ J kg}^{-1} \text{ K}^{-1}$                                                                                            | <b>A1</b> |
| 2(c)      | same gain in internal energy so same temperature rise                                                                               | <b>B1</b> |
|           | no change in volume so no work done<br><b>or</b><br>no work done so less thermal energy needed (for same change in internal energy) | <b>B1</b> |
|           | less thermal energy needed (for same temperature change) so lower specific heat capacity                                            | <b>B1</b> |

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|----------|----------------------------------|-------|
| 2(c)(i)  | Number of molecules (of the gas) | D1    |